1 General Math

Identities

- $|\mathbf{v}-\mathbf{u}|^2 = |\mathbf{u}|^2 + |\mathbf{v}|^2 - 2\,|\mathbf{u}|\,|\mathbf{v}|\cos\theta$
- $\sec^2\theta = 1+\tan^2\theta, \qquad \csc^2\theta = 1+\cot^2\theta$
- $\sin(\alpha\pm\beta) = \sin\alpha\cos\beta \pm \cos\alpha\sin\beta$
- $\cos(\alpha\pm\beta) = \cos\alpha\cos\beta \mp \sin\alpha\sin\beta$
- $\tan(\alpha\pm\beta) = \frac{\tan\alpha\pm\tan\beta}{1\mp\tan\alpha\tan\beta}$
- $\cos^2\theta = \frac{1}{2}(1+\cos(2\theta))$

Coordinate Systems

In the **Cartesian coordinate system**, we use the right hand rule to identify the x, y, z axes (Index towards the x, middle towards the y, thumb towards the z).

We change to the other coordinate systems with:

$$r=\sqrt{x^2+y^2}, \quad \tan\theta=\frac{y}{x}, \quad z=z.$$
$$\rho=\sqrt{x^2+y^2+z^2}, \quad \cos\phi=\frac{z}{\rho}, \quad \tan\theta=\frac{y}{x}.$$

In the **Cylindrical coordinate system**, we represent each point P by (r,θ,z), such that $r\geq 0$ is the length the projection of $\overrightarrow{OP}$ on the xy-plane and θ its angle with the x-axis. We change to the other corrdinate systems with:

$$x=r\cos\theta, \quad y=r\cos\theta, \quad z=z.$$
$$\rho=\sqrt{r^2+z^2}, \quad \cos\phi=\frac{z}{\rho}, \quad \theta=\theta.$$

In the **Spherical coordinate system**, we represent each point P by (ρ,ϕ,θ) such that ρ is the distance from the origin, $\phi\in[0,\pi]$ is the angle between $\overrightarrow{OP}$ and the z-axis. We change to the other coordinate systems with:

$$r=\rho\sin\phi, \quad \theta=\theta, \quad z=\rho\sin\phi.$$
$$x=\rho\sin\phi\cos\theta, \quad y=\rho\sin\phi\sin\theta \quad z=\rho\sin\phi.$$

Vector Products

The **dot product** of two vectors $\mathbf{u}, \mathbf{v}$ in $\mathbb{R}^3$ is defined:

$$\mathbf{u}\cdot\mathbf{v} = u_1v_1+u_2v_2+u_3v_3 = |u|\,|v|\cos\theta$$

where θ is the angle between $\mathbf{u}$ and $\mathbf{v}$.

- $\mathbf{u}, \mathbf{v}$ are orthogonal $\iff \mathbf{u}\cdot\mathbf{v}=0$.
- $\forall\mathbf{v}$, $(\mathbf{0},\mathbf{v})$ are orthogonal.
- The projection of $\mathbf{u}$ in the direction of $\mathbf{v}$ is defined:
$$\operatorname{proj}_{\mathbf{v}}\mathbf{u} = \left(\frac{\mathbf{u}\cdot\mathbf{v}}{|\mathbf{v}|}\right)\hat{\mathbf{v}}.$$
The scalar component can be negative, when $\theta\in\ ]\frac{\pi}{2},\pi]$

The **cross product** of two vectors $\mathbf{b}=\langle b_1,b_2,b_3\rangle$ and $\mathbf{c}=\langle c_1,c_2,c_3\rangle$ is defined:

$$\mathbf{b}\times\mathbf{c} = \begin{vmatrix}\hat{\imath} & \hat{\jmath} & \hat{k}\\ b_1 & b_2 & b_3\\ c_1 & c_2 & c_3\end{vmatrix} = |\mathbf{u}|\,|\mathbf{v}|\,(\sin\theta)\,\mathbf{n}.$$

where θ is the angle between $\mathbf{u}$ and $\mathbf{v}$ and $\mathbf{n}$ is the unit vector in the direction according to the right hand rule.

- $\forall\mathbf{a}\in\mathbb{R}^3, \mathbf{a}\cdot(\mathbf{b}\times\mathbf{c}) = \begin{vmatrix}a_1 & a_2 & a_3\\ b_1 & b_2 & b_3\\ c_1 & c_2 & c_3\end{vmatrix} = (\mathbf{b}\times\mathbf{c})\cdot\mathbf{a}$
- Cross product is defined in $\mathbb{R}^2$ by taking them as vectors on the xy-plane of $\mathbb{R}^3$.
- $\mathbf{u},\mathbf{v}$ are parallel $\iff \mathbf{u}\times\mathbf{v}=\mathbf{0}$
- The cross product is the area of the parallelogram.
- $|\mathbf{u}\times\mathbf{v}| = |\mathbf{u}|\,|\mathbf{v}|\sin\theta$

2 "Pre-Calculus"

Lines

Given a point S, its distance to the line L is given by: $\frac{|\overrightarrow{PS}\times\mathbf{v}|}{|\mathbf{v}|}$

Planes

To find the distance from a point from a point to a plane, find a vector from the plane to the point $\mathbf{v}$, and take the dot product of $\mathbf{v}$ with $\mathbf{n}$ to find the scalar multiple of $\mathbf{n}$.

$$\text{Distance} = \left|\frac{\mathbf{u}\cdot\mathbf{v}}{|\mathbf{v}|}\right|.$$

Cylinders

Given a curve $z=f(y)$ on the yz-plane, "moved" along the x-axis, we have points $(x,y,f(y))$ on the cylinder generated by the curve, where x is arbitrary. The surface has the equation $z=f(y)$. The other planes and axes can also be used.

3 Vector Calculus

We call functions $\mathbf{r}:\mathbb{R}\to\mathbb{R}^3$ a **vector function** or a **vector-valued function**. We may write $\mathbf{r}$ as:

$\mathbf{r}(t) = \langle f(t), g(t), h(t)\rangle = f(t)\,\hat{\imath}+g(t)\,\hat{\jmath}+h(t)\,\hat{k}$.

We may interpret $\mathbf{r}(t)$ as a point $P=(f(t),g(t),h(t))$ moving in space over time t. We form a curve in space with its path.

Limits

For vector function $\mathbf{r}(t)=f(t)\,\hat{\imath}+g(t)\,\hat{\jmath}+h(t)\,\hat{k}$ and vector $\mathbf{L}$, $\lim_{t\to t_0}\mathbf{r}(t)=\mathbf{L} \iff \forall\epsilon>0\exists\delta>0$ such that

$$\forall t, 0<|t-t_0|<\delta \implies |\mathbf{r}(t)-\mathbf{L}|<\epsilon.$$

It can be shown that the limit of a vector function comprises the limits of its coordinates:

$$\lim_{t\to t_0}\mathbf{r}(t) = \left\langle \lim_{t\to t_0}f(t), \lim_{t\to t_0}g(t), \lim_{t\to t_0}h(t)\right\rangle.$$

The vector function is continuous at a point $t=t_0$ in its domain if

$$\lim_{t\to t_0}\mathbf{r}(t)=\mathbf{r}(t)$$

and it is a continuous function $\iff$ it is continuous at every point in its domain.

Differentiation

We define the derivative of a vector function as

$$\mathbf{r}'(t) = \frac{d\mathbf{r}}{dt} = \lim_{\Delta t\to 0}\frac{\mathbf{r}(t+\Delta t)-\mathbf{r}(t)}{\Delta t}.$$
$$\mathbf{r}'(t) = \frac{df}{dt}\hat{\imath}+\frac{dg}{dt}\hat{\jmath}+\frac{dh}{dt}\hat{k}.$$

The derivative does not exist if the limit does not exist. We may interpret a nonzero derivative as a **tangent vector** to the curve at the given t. We then have the tangent line that which is spanned by $\mathbf{r}'(t)$.

If $\forall t, \mathbf{r}'(t)$ exists, $\mathbf{r}(t)$ is differentiable. If $\mathbf{r}(t)$ is differentiable (continuous) and $\forall t, \mathbf{r}'(t)\neq\mathbf{0}$,the curve is smooth.

We may also interpret derivatives as the **velocity** of the point moving in space over time. That is, $\mathbf{v}=\frac{d\mathbf{r}}{dt}$, speed$=|\mathbf{v}|$, acceleration $\mathbf{a}=\frac{d\mathbf{v}}{dt}=\frac{d^2\mathbf{r}}{dt^2}$, and the direction of motion at time t is $\frac{\mathbf{v}}{|\mathbf{v}|}$

Integration

We define the **indefinite integral** $\int\mathbf{r}\,dt$ the set of all anti-derivatives of $\mathbf{r}$.

$$\int\mathbf{r}\,dt = \left\langle\int f(t)\,dt, \int g(t)\,dt, \int h(t)\,dt\right\rangle.$$

The **definite integral** is defined, given integrable components over the interval $[a,b]$: $\int_a^b\mathbf{r}\,dt=$

$$\left(\int_a^b f(t)\,dt\right)\hat{\imath}+\left(\int_a^b g(t)\,dt\right)\hat{\jmath}+\left(\int_a^b h(t)\,dt\right)\hat{k}$$

Application: Projectile Motion

We have the following results from application of vector calculus on projectile motion, thrown at an angle α above the x-axis with initial velocity $\mathbf{v}_0$, and $v_0=|\mathbf{v}_0|$:

- Horizontal displacement $x=tv_0\cos\alpha$
- Vertical acceleration $y''=-g$
- Vertical velocity $\frac{dy}{dt}=-gt+C_1=v_0\sin\alpha-gt$
- Vertical displacement $y=tv_0\sin\alpha-\frac{1}{2}gt^2+C_2$
- Displacement $\mathbf{r}=\left\langle rv_0\cos\alpha, tv_0\sin\alpha-\frac{1}{2}gt^2\right\rangle$
- Velocity $\mathbf{v}=\langle v_0\cos\alpha, v_0\sin\alpha-gt\rangle$
- Maximum height at $y'=0, tmax=\frac{v_0\sin\alpha}{g}$
- $y=0$ at $t=0$ or $t=\frac{2v_0\sin\alpha}{g}$

Arc Lengths

To motivate the derivation of arc length, we consider a short straight segment of the curve

$$\Delta L \approx \sqrt{(\Delta x)^2+(\Delta y)^2+(\Delta z)^2} = \sqrt{\left(\frac{dx}{dt}\right)^2+\left(\frac{dy}{dt}\right)^2+\left(\frac{dz}{dt}\right)^2}\Delta t$$

Taking the sum of the small arc segments, we obtain the length of the smooth curve $\mathbf{r}(t)=\langle x(t),y(t),z(t)\rangle, \quad a\leq t\leq b$ as t increases from $t=a$ to $t=b$ is:

$$L=\int_a^b\sqrt{\left(\frac{dx}{dt}\right)^2+\left(\frac{dy}{dt}\right)^2+\left(\frac{dz}{dt}\right)^2}\,dt.$$

We call L the arc length of the curve.

Considering a function $s(t)=\int_{t_0}^t|\mathbf{v}(t)|\,dt$ the arc length from t_0 to t, we have the unit tangent vector of the smooth curve

$$\mathbf{T}(t)=\frac{\mathbf{v}(t)}{|\mathbf{v}(t)|}=\frac{d\mathbf{r}(s)}{ds}=\frac{d\mathbf{r}(t)/dt}{ds/dt}.$$

Curvature

As a measure of the amount of "curving", we define the curvature of a smooth curve:

$$\kappa=\left|\frac{d\mathbf{T}(s)}{ds}\right|=\frac{1}{|\mathbf{v}|}\left|\frac{d\mathbf{T}}{dt}\right|.$$

That is, the rate of change of direction of the unit tangent vector. We may also interpret curvature as the length of the acceleration vector while travelling on the curve at $|\mathbf{v}|=1$.

4 Multivariate Calculus

We call functions $f:D\to\mathbb{R}, D\subseteq\mathbb{R}^n$ a **real-valued function** on D. f is a function of n independent variables x_1 to x_n.

$$w=f(x_1,x_2,\ldots,x_n).$$

We say w is a dependent variable of f.

Consider the set

$\{(x_1,\ldots,x_n,w):\mathbf{x}=(x_1,\ldots,x_n)\in D, w=f(\mathbf{x})\}$

This set is known as the graph of f.

Given a constant value c, consider the set

$$\{f(\mathbf{x})=c:\mathbf{x}\in\mathbb{R}^n\}$$

This set is known as the level set of f. The level set is a hypersurface for non-critical points of f. If f is continuous, the level set has to be closed.

In the case of $z=f(x,y)$, the graph is a surface of which the projection onto the xy-plane is the domain D, and the level sets are curves on the xy-plane. A level set can be interpreted as the intersection between the plane $z=c$ and the graph of f.

Interior and Boundary Points

For R a domain of f in the xy-plane

- A point is an interior point of R if it is the center of a small disk wholly contained in R.
- A point is a boundary point of R if every disk centered at the point is partially in R and partially outside R.

Limits

For n-variable function $f:D\to\mathbb{R}$, we extend the definition of **limits** for multivariable functions as follows:

$\lim_{(x,y)\to(x_0,y_0)}f(x,y)=L \iff \forall\epsilon>0\exists\delta>0$ such that $\forall(x,y)\in D\setminus\{(x_0,y_0)\}$,

$\sqrt{(x-x_0)^2+(y-y_0)^2}<\delta \iff |f(x,y)-L|<\epsilon.$

Suppose $L,M,k\in\mathbb{R}$ such that

$$\lim_{(x,y)\to(x_0,y_0)}f(x,y)=L \qquad \lim_{(x,y)\to(x_0,y_0)}g(x,y)=M.$$

and $r\in\mathbb{Z}$, $s\in\mathbb{Z}^*$ with no common factors

Properties of Limits of n-Variable functions:

- $\lim_{(x,y)\to(x_0,y_0)}(f(x,y)+g(x,y))=L+M$
- $\lim_{(x,y)\to(x_0,y_0)}(f(x,y)\cdot g(x,y))=L\cdot M$
- $\lim_{(x,y)\to(x_0,y_0)}(kf(x,y))=kL$
- $M\neq 0 \implies \lim_{(x,y)\to(x_0,y_0)}\left(\frac{f(x,y)}{g(x,y)}\right)=\frac{L}{M}$
- $L^{r/s}\in\mathbb{R} \implies \lim_{(x,y)\to(x_0,y_0)}(f(x,y))^{\frac{r}{s}}=L^{r/s}$
- f,g are polynomials in x,y and $g(a,b)\neq 0$, then
$$\lim_{(x,y)\to(a,b)}f(x,y)=f(a,b)$$
$$\lim_{(x,y)\to(a,b)}\frac{f(x,y)}{g(x,y)}=\frac{f(a,b)}{g(a,b)}.$$

Limits in 2 variables is more complicated as (x,y) could approach (x_0,y_0) from many directions.

We may compute by comparison (by "squeezing"), or conversion to other coordinate systems.

To show the non-existence of a limit, we then use the **two-path test**. If a function $f(x,y)$ has different limits along two different paths as (x,y) approaches (x_0,y_0), then the limit $\lim_{(x,y)\to(x_0,y_0)}f(x,y)$ does not exist.

4.1 Multivariate Derivatives

The partial derivatives of $f(x,y)$ wrt x, $f_x(x_0,y_0)$ is

$$\left.\frac{\partial f}{\partial x}\right|_{x_0,y_0}=\lim_{\triangle x\to 0}\frac{f(x_0+\triangle x,y_0)-f(x_0,y_0)}{\triangle x},$$

provided it exists. We may interpret a partial derivative as the gradient of the tangent line to the graph $z=f(x,y_0)$ at $x=x_0$. Similarly, we have the partial derivatives wrt y

$$\left.\frac{\partial f}{\partial y}\right|_{x_0,y_0}=\lim_{\triangle y\to 0}\frac{f(x_0,y_0+\triangle y)-f(x_0,y_0)}{\triangle y}.$$

If the first partial derivatives are defined in an open region R containing (x_0,y_0), and are continuous at (x_0,y_0), the change $\triangle z$ of f by moving from (x_0,y_0) to $(x_0+\triangle x, y_0+\triangle y)\in R$ satisfies an equation of the form

$\triangle z=f_x(x_0,y_0)\triangle x+f_y(x_0,y_0)\triangle y+\epsilon_1\triangle x+\epsilon_2\triangle y$

where $\triangle x,\triangle y\to 0 \implies \epsilon_1,\epsilon_2\to 0$.

We further define higher partial derivatives $f_{yx}=\frac{\partial^2 f}{\partial x\partial y}$. If $f(x,y)$ and its partial derivatives f_x,f_y,f_{xy},f_{yx} are defined throughout an open region containing (a,b) and are continuous at (a,b),

$$f_{xy}(a,b)=f_{yx}(a,b).$$

Directional Derivatives

The gradient of $f(x_1,\dots,x_n)$ at $P_0(a_1,\dots,a_n)$ is

$$\nabla f = \langle f_{x_1}, f_{x_2}, \dots, f_{x_n}\rangle .$$

$\nabla f(P_0)$ is perpendicular to the level set $f(P_0)=c$.

Properties of gradients:

For $k,l\in\mathbb{R}$, $\nabla(kf)\pm\nabla(lg)=k\nabla f\pm l\nabla g$.

$$\nabla(fg)=f\nabla g+g\nabla f,\quad \nabla\left(\frac{f}{g}\right)=\frac{g\nabla f-f\nabla g}{g^2}.$$

Directional derivatives generalise partial derivatives to any direction with unit vector $\mathbf{u}=u_1\hat{\imath}+u_2\hat{\jmath}$

$$\left(\frac{df}{ds}\right)_{\mathbf{u},P_0}=\lim_{s\to0}\frac{f(x_0+su_1,y_0+su_2)-f(x_0,y_0)}{s}$$
$$=\mathbf{u}\cdot\nabla f(x_0,y_0)=D_{\mathbf{u}}f(x_0,y_0).$$

The rate of increase/decrease is greatest when $\mathbf{u}$ is in the direction of $\pm\nabla f$. $D_{\mathbf{u}}f=\pm\sqrt{f_x^2+f_y^2}$. The rate of change is 0 when $\mathbf{u}$ is perpendicular to ∇f, equivalently, parallel to the level set.

Tangent Space

The tangent plane to $f(x,y,z)=c$ at $P_0(x_0,y_0,z_0)$ is

$$f_x(P_0)(x-x_0)+f_y(P_0)(y-y_0)+f_z(P_0)(z-z_0)=0.$$

In general, the tangent space is defined by the specified point, and a orthogonal vector to all directional derivatives at the point.

The tangent space is an approximation of the graph. The approximation is better in directions parallel to the axes.

Continuity

At $(c_0,c_1,\dots,c_n)$, the n-variable function $f:D\to\mathbb{R}$ is **continuous** if:

- $(c_0,c_1,\dots,c_n)\in D$,
- $L:=\lim_{(x,\dots,x_n)\to(c_0,\dots,c_n)} f(x_0,\dots,x_n)$ exists, and
- $L=f(c_0,\dots,c_n)$.

f is continuous $\iff$ f is continuous $\forall(c_0,c_n)\in D$. Equivalently, f is differentiable $\implies$ continuous at the point if:

- All first order partial derivatives exist at the point,
- The tangent space is a good approximation near the point.

f is differentiable $\iff$ f is differentiable $\forall(c_0,\dots,c_n)\in D$.

Chain Rule

For a differentiable function $w=f(x_1,\dots,x_n)$ where each $x_i=x_i(t_1,\dots,t_m)$ is differentiable,

$w(t_1,\dots,t_m)=f(x_1(t_1,\dots,t_m),\dots,x_n(t_1,\dots,t_m))$ is differentiable and

$$\frac{\partial w}{\partial t_j}=\frac{\partial f}{\partial x_1}\frac{\partial x_1}{\partial t_j}+\frac{\partial f}{\partial x_2}\frac{\partial x_2}{\partial t_j}+\dots+\frac{\partial f}{\partial x_n}\frac{\partial x_n}{\partial t_j}.$$

$$\left(\frac{\partial w}{\partial t_1},\dots\frac{\partial w}{\partial t_m}\right)=\left(\frac{\partial w}{\partial x},\dots,\frac{\partial w}{\partial x_n}\right)\begin{pmatrix}\frac{\partial x_1}{\partial t_1}&\cdots&\frac{\partial x_1}{\partial t_m}\\ \vdots&\ddots&\vdots\\ \frac{\partial x_n}{\partial t_1}&\cdots&\frac{\partial x_n}{\partial t_m}\end{pmatrix}$$

Extreme Values

For a n-variable function f defined on a region R containing P:

- $f(P)$ is a local maximum value if $f(P)>f(P')$ for all P' in an open set centred at P
- $f(P)$ is a local minimum value if $f(P)<f(P')$ for all P' in an open set centred at P
- P is a local extreme point if it is a local maximum or minimum.
- P is a saddle point if in every open set centred at P, there exist P' where $f(P)>f(P')$, P'' where $f(P)<f(P'')$.

An interior point is a critical point if one of the partial derivatives are 0 or undefined. Extrema must occur at critical or boundary points.

We define the discriminant $D=f_{xx}f_{yy}-(f_{xy})^2$ that describes the local shape of the graph. If the first and second partial derivatives are continuous in a open set centred at P and the first-order partial derivatives are 0, with the **Second Derivative Test**,

- $D<0$ at $P\implies P$ is a saddle point.
- $f_{xx}<0$ and $D>0$ at $P\implies P$ is a local maximum.
- $f_{xx}>0$ and $D>0$ at $P\implies P$ is a local minimum.
- $D=0$ at $P\implies$ The test is inconclusive.

Lagrange Multipliers

Within a constraint $g(x_1,x_2,\dots,x_n)=0$, to find the maximum value of $f(x_1,x_2,\dots,x_n)$. At the maximal point P_0, their gradients are parallel. For some Lagrange multiplier λ,

$$\nabla f(P_0)=\lambda\nabla g(P_0).$$

For $\nabla g\neq\mathbf{0}$, P_0 is the solution to the system of equations

$$\begin{cases}g(x_1,x_2,\dots,x_n)=0\\ \nabla f(x_1,x_2,\dots,x_n)=\lambda\nabla g(x_1,x_2,\dots,x_n)\end{cases}.$$

Taylor's Expansion

Consider $f(x,y)$. We define for $a,b\in\mathbb{R},n\in\mathbb{N}^*$,

$$\left(a\frac{\partial}{\partial x}+b\frac{\partial}{\partial y}\right)^n f=\sum_{k=0}^{n}\binom{n}{k}a^{n-k}b^k\frac{\partial^n f}{\partial x^{n-k}\partial y^k}.$$

The Taylor's expansion for $f(x,y)$ at the origin is

$$\begin{aligned}f(x,y)&=f+xf_x+yf_y\\&+\frac{1}{2!}\left(x^2f_{xx}+2xyf_{xy}+y^2f_{yy}\right)\\&+\frac{1}{3!}\left(x^3f_{xxx}+3x^2yf_{xxy}+3xy^2f_{xyy}+y^3f_{yyy}\right)\\&+\dots+\frac{1}{n!}\left(x\frac{\partial}{\partial x}+y\frac{\partial}{\partial y}\right)^n f+error\end{aligned}$$

with each partial derivative evaluated at $(0,0)$, and a small error term (for "nice" functions and large n)

$$error=\frac{1}{(n+1)!}\left(x\frac{\partial}{\partial x}+y\frac{\partial}{\partial y}\right)^n f\Big|_{(cx,cy)}.$$

4.2 Multivariate Integrals

The Riemann Integral of $f(x,y)$ over a region R is the limit of partitioning R into $n\to\infty$ divisions of area $\Delta A_k\to0$ for the volume of $z=f(x,y)$. That of $f(x,y,z)$ over a volume D is the limit of partitioning D each of volume ΔV_k for a summation of f over D. If this limit exists, f is *Riemann integral over* R.

Properties of Riemann Integrals:

- $\int_R c_1f\pm c_2g\,dA=c_1\int_R f\,dA\pm c_2\int_R g\,dA$
- $\forall(x,y)\in R, f>g\implies\int_R f\,dA>\int_R g\,dA$
- For a partition of $R=R_1\cup R_2$,
$$\int_R f\,dA=\int_{R_1}f\,dA+\int_{R_2}f\,dA$$

By **Fubini's theorem**, we can compute the Riemann integral of f continuous over a region R, by iterating the integration with respect to each axis.

Applications

We find the area/volume of the region by setting $f=1$ and the average value of f over R with $\frac{1}{Area\ of\ R}\iint_R f\,dA$.

The surface area of the graph consists areas $\Delta G_k=\frac{\Delta A_k}{\mathbf{n}\cdot\mathbf{k}}$, where

$$\mathbf{n}\cdot\mathbf{k}=\frac{\langle -f_x,-f_y,1\rangle}{\sqrt{1+f_x^2+f_y^2}}\cdot\langle0,0,1\rangle\implies$$
$$Area=\iint_R\sqrt{1+f_x^2+f_y^2}\,dxdy.$$

For an object occupying D in 3-dim space with density $\delta(x,y,z)$,

- Mass $M=\iiint_D\delta(x,y,z)\ dV$
- Moment about yz-plane $M_{yz}=\iiint_D x\delta(x,y,z)\ dV$
- Moment about xz-plane $M_{xz}=\iiint_D y\delta(x,y,z)\ dV$
- Moment about xy-plane $M_{xy}=\iiint_D z\delta(x,y,z)\ dV$
- Center of mass: $\bar{x}=\frac{M_{yz}}{M},\bar{y}=\frac{M_{xz}}{M},\bar{z}=\frac{M_{xy}}{M}$

Substitutions

In polar/cylindrical coordinates, $\Delta A_k=r_k\Delta r_k\theta_k$, and in spherical coordinates, $\Delta V=\rho^2\sin\phi\,\Delta\rho\Delta\phi\Delta\theta$.

$$dA=rdrd\theta,\quad dV=rdrd\theta dz=\rho^2\sin\phi\,d\rho d\phi d\theta.$$

To perform substitutions for a function F,

$$x=g(u,v,w),\quad y=h(u,v,w),\quad z=k(u,v,w)$$
$$H(u,v,w)=F(g(u,v,w),h(u,v,w),z(u,v,w))$$

The map from the uvw-coordinate space to the domain of integration $T:(u,v,w)\mapsto\langle g(u,v,w),h(u,v,w),k(u,v,w)\rangle$ must be a bijection, and the Jacobian yields the multiple

$$J(u,v,w)=\begin{vmatrix}x_u&x_v&x_w\\y_u&y_v&y_w\\z_u&z_v&z_w\end{vmatrix}$$

$$\iiint_R F(x,y,z)\,dx\,dy\,dz$$
$$=\iiint_D H(u,v,w)\,|J(u,v,w)|\ du\,dv\,dw$$

5 Vector Fields

A **vector field** on $\mathbb{R}^n$ is a function $\mathbf{F}:\mathbb{R}^n\to\mathbb{R}^n$. A gradient field is the field of gradients

$\nabla f(x,y,z)=\langle f_x(x,y,z),f_y(x,y,z),f_z(x,y,z)\rangle$.

where f is the potential function of the gradient field. The line integral along segment C is given by

$$\int_C f ds=\int_b^a f(\mathbf{r}(t))\,|\mathbf{v}(t)|\ dt,\quad\text{for some smooth }\mathbf{r}$$

The line integral is dependent on the path C, but not on the motion $\mathbf{r},\mathbf{v}$. It could be interpreted as the area under the path.

Properties of Vector Fields

Let $\mathbf{F}$ be a vector field on an open connected domain D.

If $\forall A,B\in D$, $\int_A^B\mathbf{F}\cdot d\mathbf{r}=f(B)-f(A)$ is the same over all paths from A to B, the integral is path independent in D and $\mathbf{F}$ is conservative on D. If $\mathbf{F}=\nabla f$ for some scalar function $f:\mathbb{R}^n\to\mathbb{R}$, f is a potential function for $\mathbf{F}$.

$$\begin{aligned}\mathbf{F}\text{ is a gradient field}&\iff\mathbf{F}\text{ is a conservative field}\\&\iff\oint_C\mathbf{F}\cdot d\mathbf{r}=0\\&\iff curl\,\mathbf{F}=\nabla\times\mathbf{F}=\langle0,0,0\rangle\end{aligned}$$

Applications

The work done by a force $\mathbf{F}$ over a smooth curve $\mathbf{r}(t)$ is

$$W=\int_{t=a}^{t=b}\mathbf{F}\cdot\mathbf{T}\,ds=\mathbf{F}(\mathbf{r}(t))\cdot\mathbf{r}'(t)\,dt.$$

The work integral is independent of the motion $\mathbf{r}$, but is path dependent.

The flow along a curve for some smooth $\mathbf{r}$ in a continuous velocity field $\mathbf{F}$ is

$$Flow=\int_a^b\mathbf{F}\cdot\mathbf{T}\,ds.$$

If the curve is a closed loop, the flow integral is the circulation about the curve. Mathematically, it is identical to the work done.

Green's Theorem

A closed curve is one whose starting point $=$ end point. A simple curve is one that does not cross itself. A directed/oriented closed curve is one with a direction. An oriented simple closed curve moving in an anti-clockwise direction has a positive orientation, and negative orientation otherwise.

For a positive oriented closed curve C, R its enclosed region, and $\mathbf{F}(x,y)=\langle M(x,y),N(x,y)\rangle$ a differentiable vector field on the xy-plane

$$\oint_C M\,dx+N\,dy=\iint_R\frac{\partial N}{\partial x}-\frac{\partial M}{\partial y}\,dx\,dy.$$
$$\oint_C M\,dy-N\,dx=\iint_R\frac{\partial M}{\partial x}+\frac{\partial N}{\partial y}\,dx\,dy.$$

For vector fields undefined at the origin, we consider a curve C_h clockwise around the origin of radius h.

Surface Integrals

The area of surface S $f(x,y,z)=c$ over a closed and bounded plane region R is, where $\mathbf{p}$ is a the unit vector normal to R,

$$Surface\ Area=\iint_R\frac{|\nabla f|}{|\nabla f\cdot\mathbf{p}|}dA.$$

For a continuous g defined at points of S, the surface integral is

$$\iint_S g d\sigma=\iint_R g(x,y,z)\frac{|\nabla f|}{|\nabla f\cdot\mathbf{p}|}dA.$$

Intuitively, an orientation of a surface is defining a unit normal vector $\mathbf{n}(p)$ for every point p on a surface, such that the vector changes continuously wrt. p. A surface with an orientation is orientable. A connected surface has either 0 or 2 orientations, and closed surfaces are orientable.

The flux of a vector field $\mathbf{F}$ across an oriented surface S in the direction of $\mathbf{n}$ is

$$Flux=\iint_S\mathbf{F}\cdot d\mathbf{S}=\iint_S\mathbf{F}(x,y,z)\cdot\mathbf{n}\,d\sigma.$$

Intuitively, it is the "amount of flowing" across a surface. For a surface S defined by $g(x,y,z)=0$,

$$Flux=\iint_R\mathbf{F}\cdot\frac{\nabla g}{|\nabla g\cdot\langle0,0,1\rangle|}dx\,dy$$

Stokes' Theorem

We define the curl the "amount of spinning" in a vector field. We use the right-hand rule for the direction of the curl.

$$curl\mathbf{F}=\nabla\times\mathbf{F}=\left\langle\frac{\partial P}{\partial y}-\frac{\partial N}{\partial z},\frac{\partial M}{\partial z}-\frac{\partial P}{\partial x},\frac{\partial N}{\partial x}-\right.$$

By Green's theorem, $\oint_C\mathbf{C}\cdot d\mathbf{r}=\iint_R curl\ \mathbf{F}\cdot\mathbf{k}dA$ The circulation (flow integral) a vector field $\mathbf{F}=\langle M,N,P\rangle$ around the boundary C of an oriented surface S with $\mathbf{n}$ in the counter-clockwise direction

$$\oint_C\mathbf{F}\cdot d\mathbf{r}=\iint_S\nabla\times\mathbf{F}\cdot\mathbf{n}d\sigma.$$

Divergence Theorem

We define the divergence the "amount of expansion/production" in a vector field.

$$div\ \mathbf{F}=\nabla\cdot\mathbf{F}=\frac{\partial M}{\partial x}+\frac{\partial N}{\partial y}+\frac{\partial P}{\partial z}.$$

The field is incompressible if $div=0$ at all points. The theorem states

$$\iint_S\mathbf{F}\cdot\mathbf{n}\,d\sigma=\iiint_D div\ \mathbf{F}dV$$